

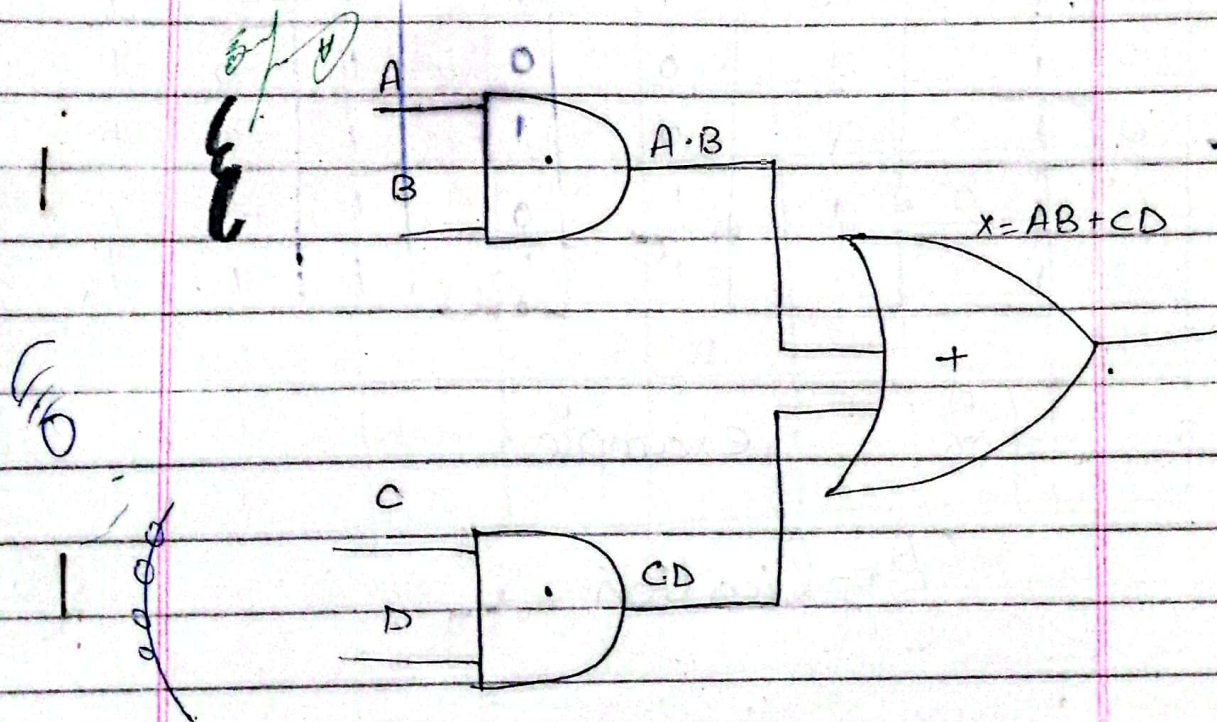
Unit #5

Combinational

Logic Circuit

It has no memory element.
→ These are the electronics circuits which consist of different electronic components (gates, transistors, ICs etc) and are used to perform different logical operations. The output depends upon the present inputs not previous.

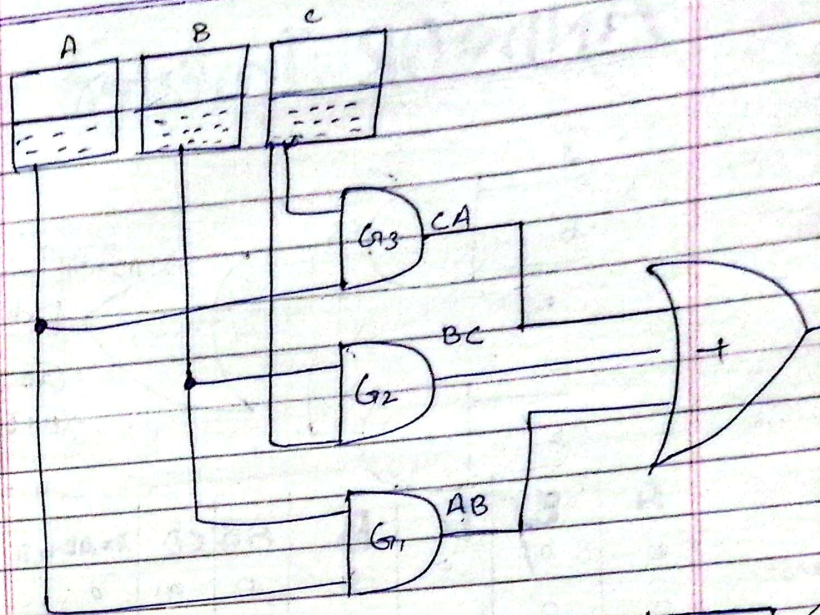
AND-OR Circuits



no. of inputs combination = $2^n = 16$

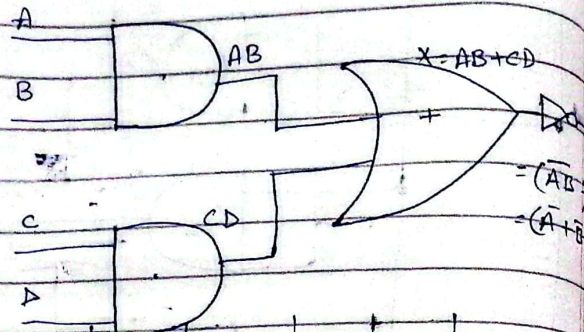
A	B	C	D	AB	CD
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	0	1	0	0
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	0	1
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	1	0
1	1	1	1	1	1

for example
Application



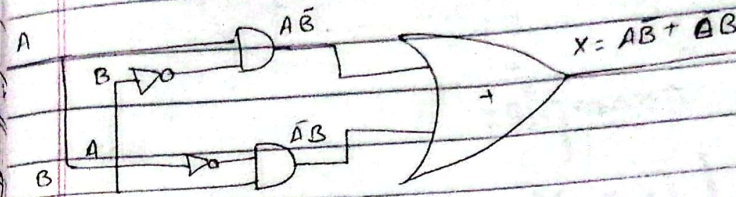
A	B	C	AB	BC	CA	X
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	1	0	1
1	0	0	0	0	0	0
1	0	1	0	0	1	1
1	1	0	1	0	0	1
1	1	1	1	1	1	1

AND-OR Inverter

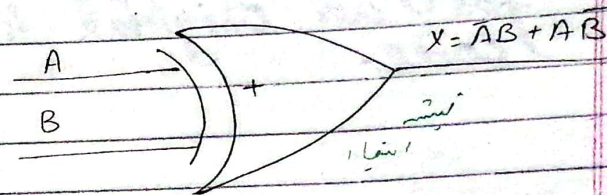


A	B	C	D	AB	CD	$X = AB + CD$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	1
0	0	0	1	0	0	0
0	0	1	1	0	0	0
0	1	0	1	0	0	0
0	1	1	1	0	1	1
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	0	0	0
1	1	1	0	1	0	1
1	0	0	1	1	0	1
1	0	1	1	1	0	1
1	1	0	1	1	0	1
1	1	1	1	1	1	1

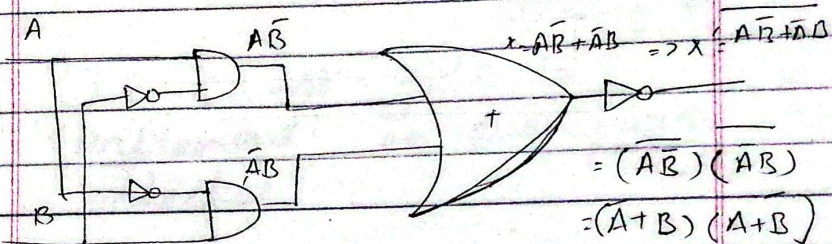
Exclusive OR



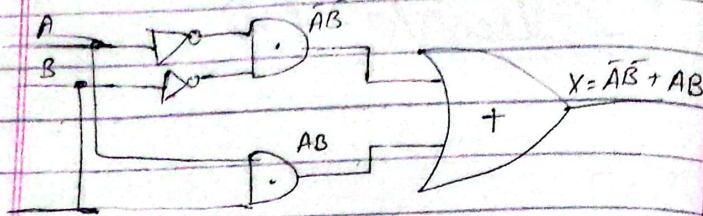
Short hand Symbol



Exclusive-NOR

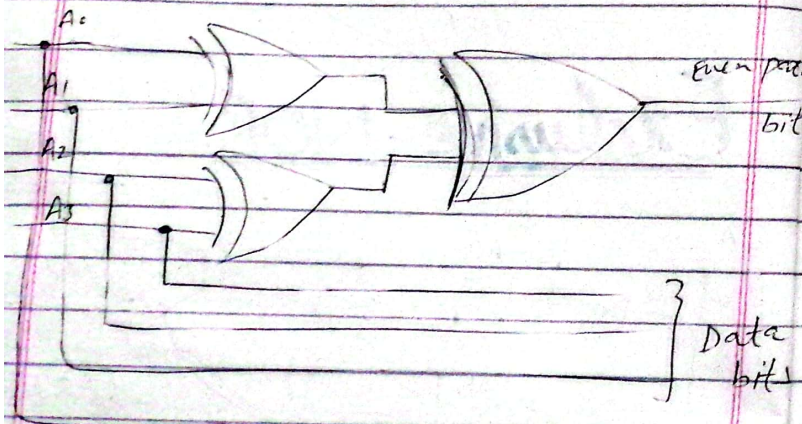


$$(\overline{A+B})(\overline{A+B}) = \overline{AB} + AB$$



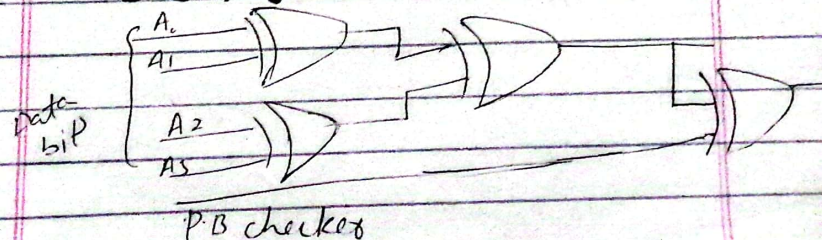
example:

Use X-OR gate to implement even parity code generator for an original 4-bit code.



A_0	A_1	A_2	A_3	P-R
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

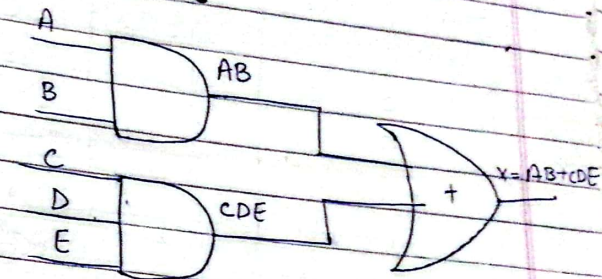
Use ex-or gate to implement an even parity checker.



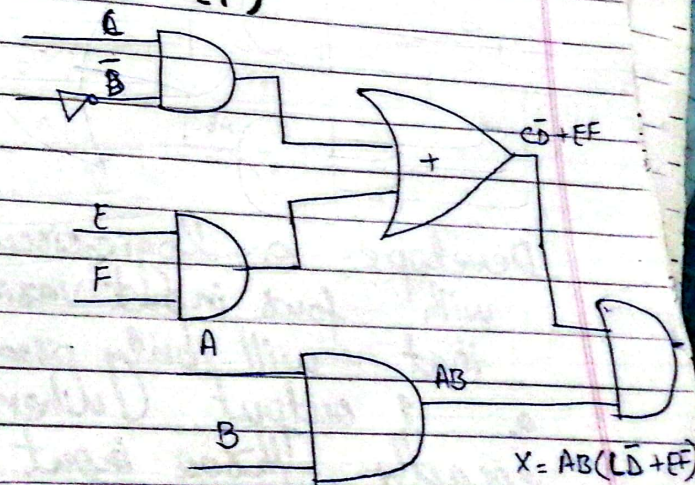
X	A_1	A_2	A_3	AB	X	D
0	0	0	0	0	0	
0	0	0	1	1	0	Error
0	0	1	0	1	0	Error
0	0	1	1	0	0	
0	1	0	0	1	0	Error
0	1	0	1	0	0	
0	1	1	0	0	0	
0	1	1	1	1	0	
1	0	0	0	1	0	Error
1	0	0	1	0	0	Error
1	0	1	0	0	0	
1	0	1	1	1	0	
1	1	0	0	0	0	Error
1	1	0	1	1	0	
1	1	1	0	1	0	Error
1	1	1	1	0	0	Error
1	1	1	1	0	0	

Implementing Combinational Logic

$$X = AB + CDE$$

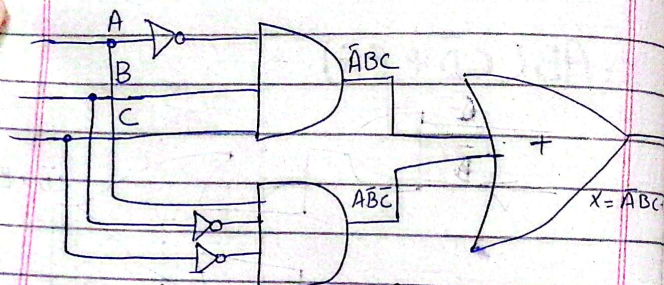


$$X = AB(C\bar{D} + EF)$$



Use table to a logic circuit.

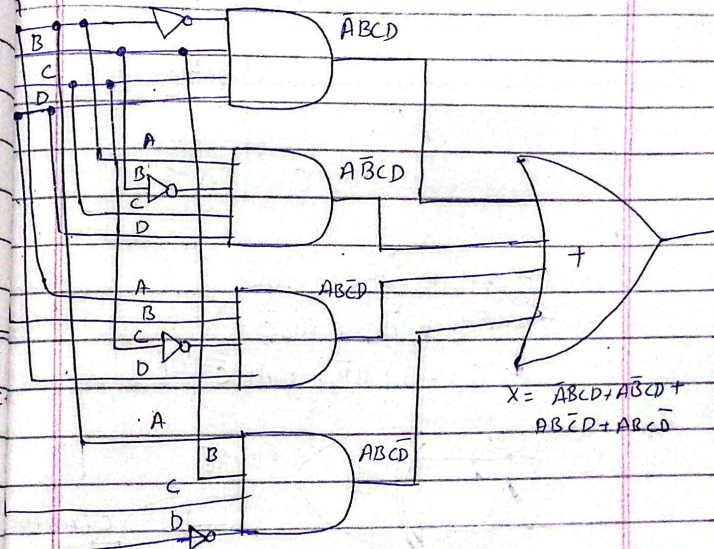
A	B	C	output	Product term
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\bar{A}BC$
1	0	0	1	$A\bar{B}\bar{C}$
1	0	1	0	
1	1	0	0	
1	1	1	0	



Develop a logic circuit with four input variables that will only produce a 1 output when exactly three input variables are 1's.

A	B	C	D	Product term
0	1	1	1	$\bar{A}BCD$
1	0	1	1	$A\bar{B}CD$
1	1	0	1	$AB\bar{C}D$
1	1	1	0	$ABCD\bar{D}$

$$X = \bar{A}BCD + A\bar{B}CD + AB\bar{C}D + ABC\bar{D}$$

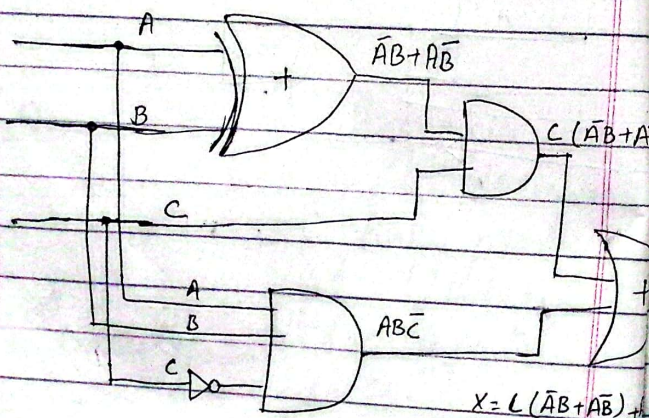


$$X = \bar{A}BCD + A\bar{B}CD + AB\bar{C}D + ABC\bar{D}$$

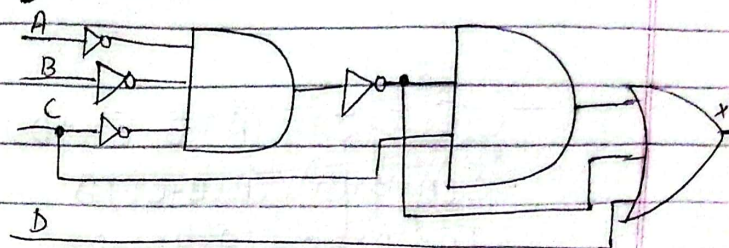
Design a logic circuit to implement the operation specified in the truth table

A	B	C	X	Product Term
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\bar{A}BC$
1	0	0	0	
1	0	1	1	$A\bar{B}C$
1	1	0	1	$AB\bar{C}$
1	1	1	0	

$$\begin{aligned}
 X &= \bar{A}BC + A\bar{B}C + AB\bar{C} \\
 &= \bar{A}BC + A\bar{B}C + AB\bar{C} \\
 &= C(\bar{A}B + A\bar{B}) + AB\bar{C}
 \end{aligned}$$



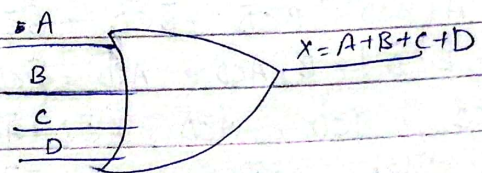
Reduce the combinational logic circuit to a minimum form.



$$X = (\bar{A}\bar{B}\bar{C})C + \bar{A}\bar{B}\bar{C} + D$$

Apply de Morgan's law

$$\begin{aligned}
 X &= (\bar{A} + \bar{B} + \bar{C})C + (\bar{A} + \bar{B} + \bar{C}) + D \\
 &= (A + B + C)C + (A + B + C) + D \\
 &= AC + BC + C + A + B + C + D \\
 &= C(A + B + 1) + A + B + D \\
 &= A + B + C + D
 \end{aligned}$$



Verify the minimized expression $A+B+C+D$ using K-map.

$$\begin{aligned}
 X &= C(\overline{A}\overline{B}\overline{C}) + (\overline{A}\overline{B}\overline{C}) + D \\
 &= C(\overline{A} + \overline{B} + \overline{C}) + (\overline{A} + \overline{B} + \overline{C}) + D \\
 &= C(A+B+C) + (A+B+C) + D \\
 &= AC + BC + C + A + B + C + D \\
 &= C(A+B+1) + A + B + D \\
 &= C + A + B + D
 \end{aligned}$$

$$\begin{aligned}
 &= C(A+\overline{A})(B+\overline{B})(D+\overline{D}) + A(B+\overline{B})(C+\overline{C})(D+\overline{D}) + B(A+\overline{A})(C+\overline{C})(D+\overline{D}) + D(A+\overline{A})(B+\overline{B})(C+\overline{C}) \\
 &= A[(B+\overline{B})(CD + C\overline{D} + \overline{C}D + \overline{C}\overline{D})] + B[(C+\overline{C})(AD + A\overline{D} + \overline{A}D + \overline{A}\overline{D})] + \\
 &\quad A[(A+\overline{A})(BD + B\overline{D} + \overline{B}D + \overline{B}\overline{D})] + D[(A+\overline{A})(BC + B\overline{C} + \overline{B}C + \overline{B}\overline{C})] \\
 &= A[BCD + BC\overline{D} + \overline{B}CD + \overline{B}\overline{C}D + \overline{B}\overline{C}\overline{D}] + B[ACD + A\overline{C}D + \overline{A}CD + \overline{A}\overline{C}D + \overline{A}\overline{C}\overline{D}] \\
 &\quad + C[ABD + AB\overline{D} + A\overline{B}D + A\overline{B}\overline{D} + \overline{A}BD + \overline{A}B\overline{D} + \overline{A}\overline{B}D + \overline{A}\overline{B}\overline{D}] + D[ABC + AB\overline{C} + A\overline{B}C + \overline{A}BC + \overline{A}\overline{B}C + \overline{A}\overline{B}\overline{C}]
 \end{aligned}$$

$$\begin{aligned}
 &= ABCD + ABC\overline{D} + A\overline{B}CD + A\overline{B}\overline{C}D \\
 &\quad + A\overline{B}\overline{C}\overline{D} + \overline{A}BCD + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D} + \overline{A}\overline{B}CD \\
 &\quad + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D}
 \end{aligned}$$

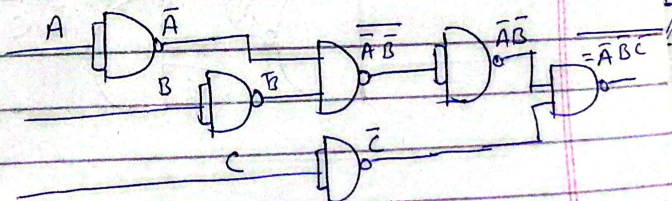
AB \ CD	00	01	11	10
00		1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$X = A + B + C + D$$

Using NAND gate logic to implement the following boolean fun.

$$X = A + B + C$$

$$\begin{aligned}
 \overline{X} &= \overline{A+B+C} \\
 &= \overline{A}\overline{B}\overline{C}
 \end{aligned}$$



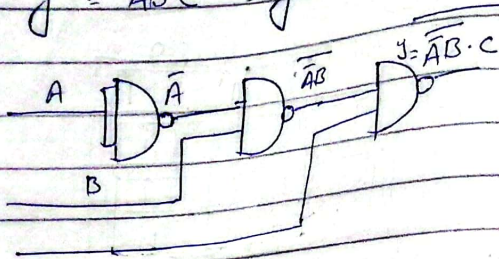
$$y = \overline{AB} + \overline{C}$$

$$\overline{y} = \overline{\overline{AB} + \overline{C}}$$

$$= \overline{\overline{AB}} \cdot \overline{\overline{C}}$$

$$\overline{y} = \overline{\overline{AB}} \cdot C$$

$$\overline{\overline{y}} = \overline{\overline{\overline{AB} \cdot C}} = y$$



$$y = (\overline{A} + \overline{B})C$$

$$\overline{y} = \overline{(\overline{A} + \overline{B})C}$$

$$\overline{y} = \overline{(\overline{A} + \overline{B})} + \overline{C}$$

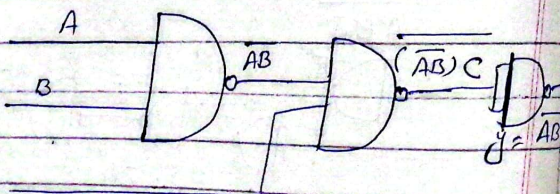
$$\overline{y} = (\overline{\overline{A} + \overline{B}}) + \overline{C}$$

$$= (\overline{\overline{A}} \cdot \overline{\overline{B}}) + \overline{C}$$

$$= (AB) + \overline{C}$$

$$= \overline{\overline{AB} \cdot \overline{C}}$$

$$= \overline{(\overline{AB})C}$$



$$y = \overline{AB(C+D)}$$

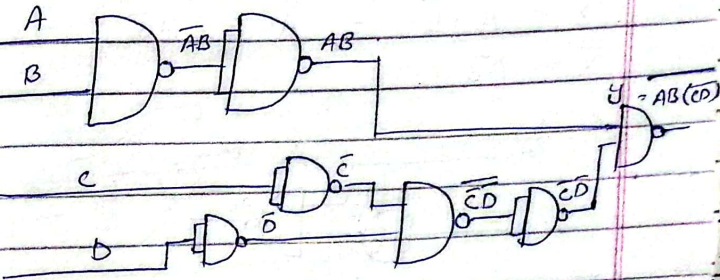
$$\overline{y} = \overline{\overline{AB} + (\overline{C+D})}$$

$$\overline{y} = \overline{\overline{AB}} + \overline{\overline{C+D}}$$

$$\overline{y} = \overline{\overline{AB}} \cdot (\overline{\overline{C+D}})$$

$$\overline{y} = \overline{\overline{AB}} \cdot (\overline{C+D})$$

$$\overline{\overline{y}} = \overline{\overline{\overline{\overline{AB}} \cdot (\overline{C+D})}} = y$$



$$y = A(\overline{AB} + C)$$

$$\overline{y} = \overline{A(\overline{AB} + C)}$$

$$\overline{y} = \overline{A} + \overline{(\overline{AB} + C)}$$

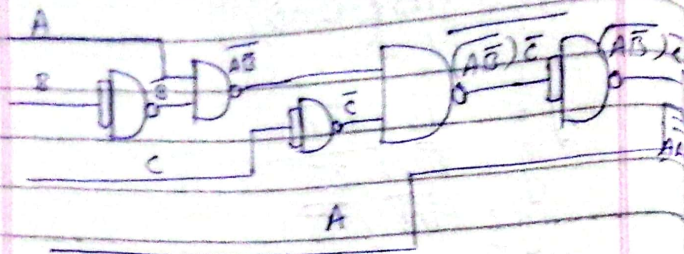
$$= \overline{A} + (\overline{\overline{AB}} \cdot \overline{C})$$

$$\overline{y} = \overline{A} + (\overline{AB} \cdot \overline{C})$$

$$\overline{y} = \overline{A} \cdot (\overline{AB} \cdot \overline{C})$$

$$\overline{\overline{y}} = \overline{\overline{\overline{A} \cdot (\overline{AB} \cdot \overline{C})}} = y$$

$$\overline{\overline{y}} = \overline{\overline{A} \cdot (\overline{AB} \cdot \overline{C})} = y$$



$$X = \bar{A} + B$$

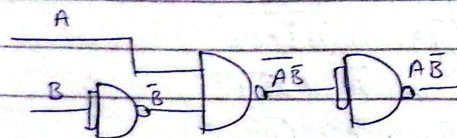
$$X = \bar{A} + B$$

$$\bar{X} = \overline{\bar{A} + B} = \bar{\bar{A}} \cdot \bar{B} = A \cdot \bar{B}$$

$$\bar{X} = A\bar{B}$$



$$X = A\bar{B}$$



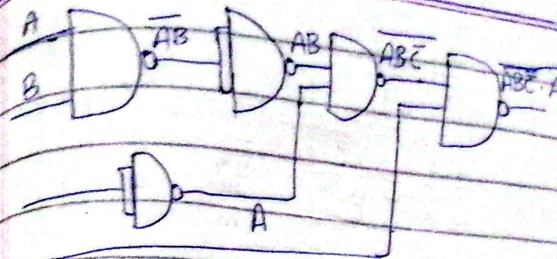
$$X = AB\bar{C} + \bar{A}$$

$$X = AB\bar{C} + \bar{A}$$

$$\bar{X} = \overline{AB\bar{C} + \bar{A}}$$

$$= \overline{AB\bar{C}} \cdot \bar{\bar{A}}$$

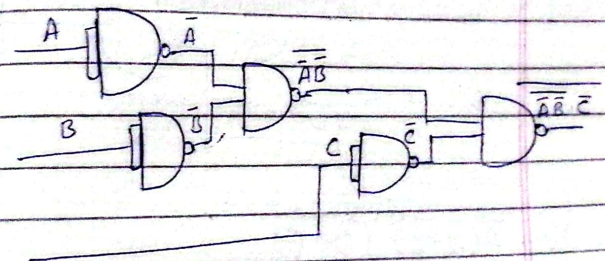
$$= \overline{AB\bar{C}} \cdot A$$



$$X = \bar{A}\bar{B} + C$$

$$\bar{X} = \overline{\bar{A}\bar{B} + C}$$

$$= \overline{\bar{A}\bar{B}} \cdot \bar{C}$$

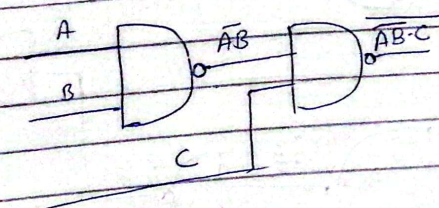


$$X = AB + \bar{C}$$

$$\bar{X} = \overline{AB + \bar{C}}$$

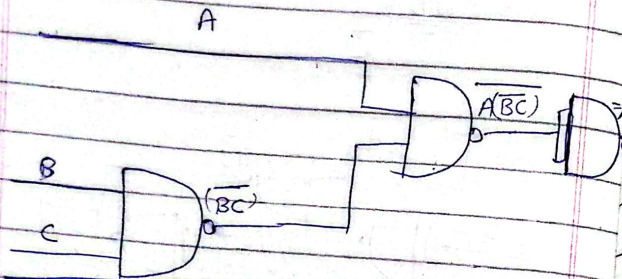
$$= \overline{AB} \cdot \bar{\bar{C}}$$

$$= \overline{AB} \cdot C$$

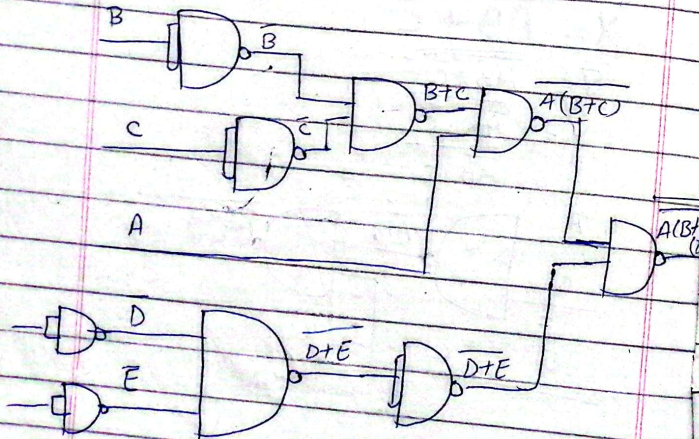


$$X = A(\bar{B} + \bar{C})$$

$$\begin{aligned}\bar{X} &= \overline{A(\bar{B} + \bar{C})} \\ &= \bar{A} + (\overline{\bar{B} + \bar{C}}) \\ &= \bar{A} + \bar{\bar{B}} \cdot \bar{\bar{C}} \\ &= \bar{A} + BC \\ &= \bar{A} \cdot \overline{BC} \\ &= A\bar{B}\bar{C}\end{aligned}$$

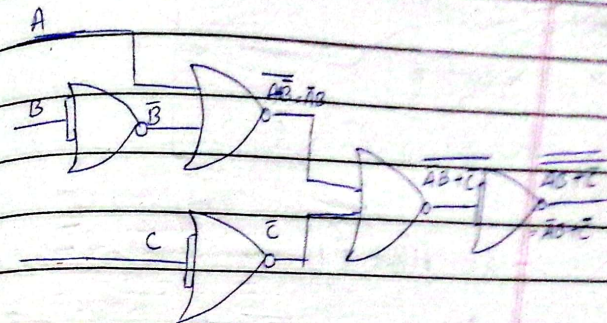


$$X = A(B+C) \cdot (\bar{D} + \bar{E})$$



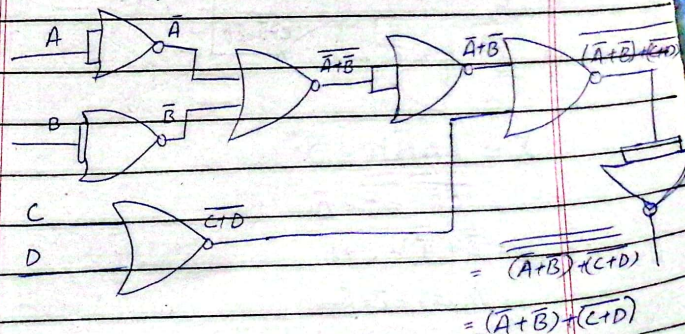
NOR Gate Logic

$$Y = \bar{A}B + \bar{C}$$



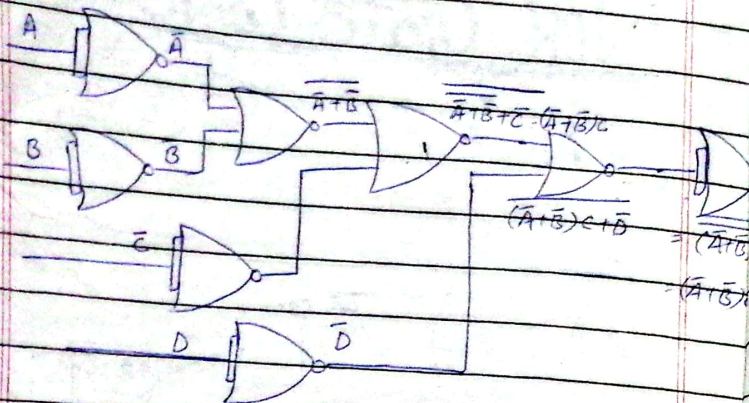
$$Y = \overline{AB(C+D)}$$

$$\begin{aligned}Y &= \overline{AB(C+D)} \\ &= (\bar{A} + \bar{B}) + (\bar{C} + \bar{D})\end{aligned}$$

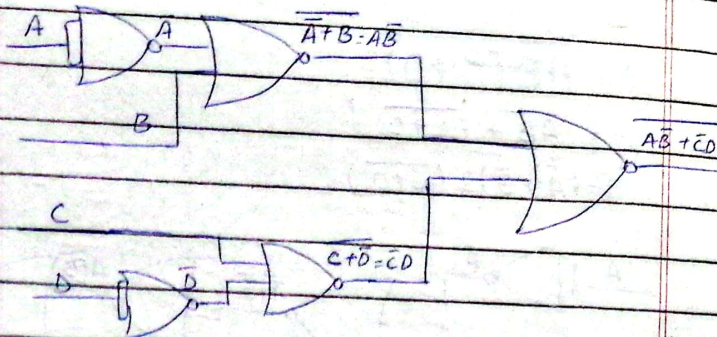


Day **MTWTF**

$$Y = (\bar{A} + \bar{B})C + \bar{D}$$

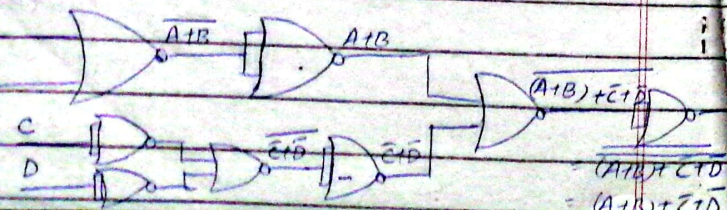


$$X = \bar{A}\bar{B} + \bar{C}D$$



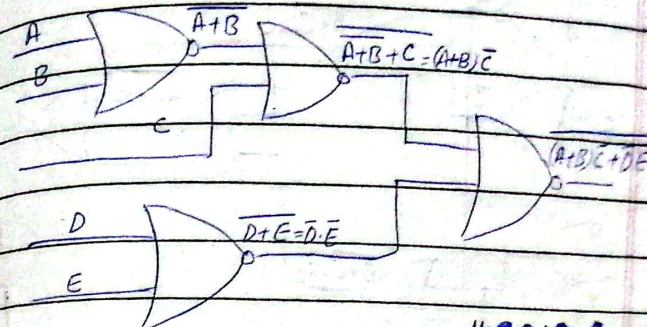
$$X = (\bar{A}\bar{B})C + \bar{D}$$

$$\begin{aligned} &= \bar{A}\bar{B} + \bar{C} + \bar{D} \\ &= (\bar{A} + \bar{B}) + \bar{C} + \bar{D} \\ &= (A+B) + \bar{C} + \bar{D} \end{aligned}$$



Day **MTWTF**

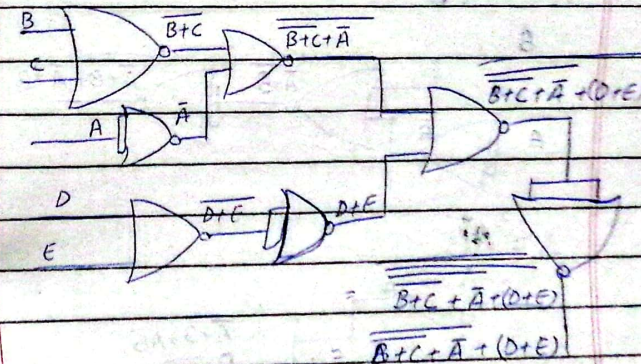
$$X = (A+B)\bar{C} + \bar{D}E$$



#2018 (PP)

$$X = \overline{A(B+C)(D+E)}$$

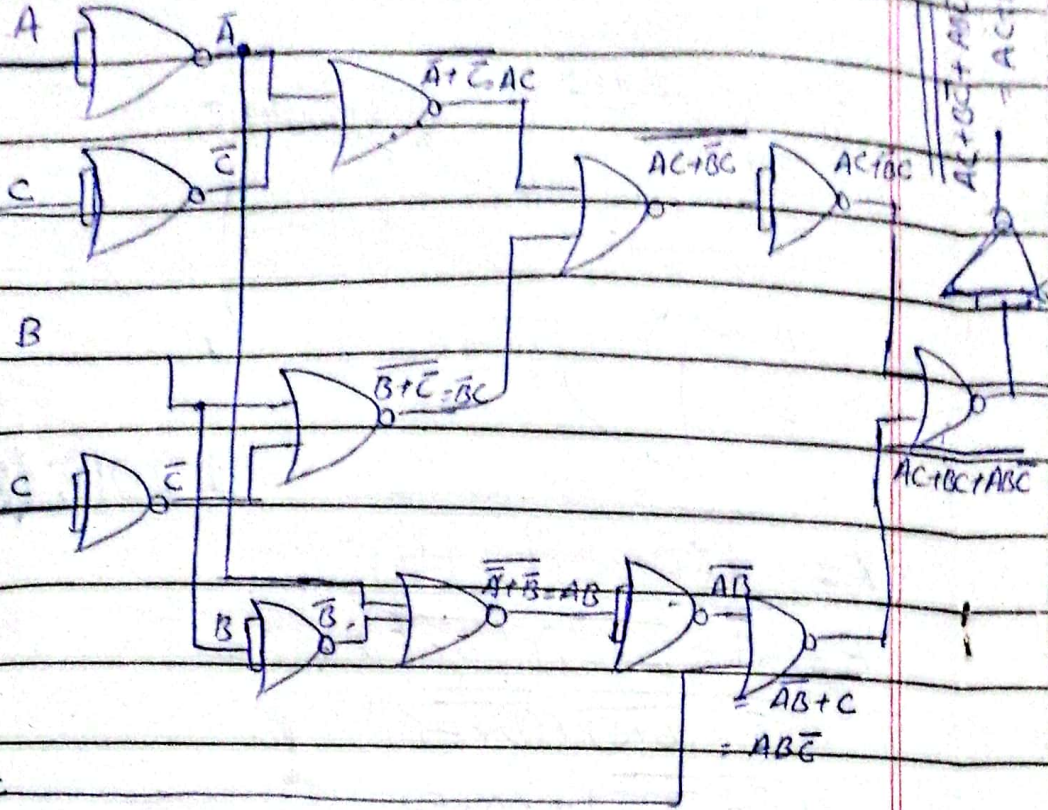
$$\begin{aligned} F &= \overline{(\bar{A} + (\bar{B} + \bar{C}))(\bar{D} + \bar{E})} \\ &= \bar{A} + (\bar{B} + \bar{C}) + (\bar{D} + \bar{E}) \\ &= (\bar{A} + \bar{B} + \bar{C}) + (\bar{D} + \bar{E}) \end{aligned}$$



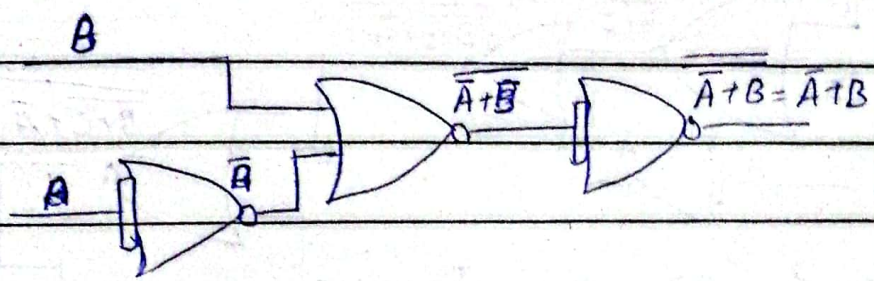
Date _____

Day **M T W T F S S**

$$y = AC + \bar{B}C + AB\bar{C}$$



$$X = \bar{A} + B$$



$$X = A\bar{B}$$

